

SOC NETWORK TRAFFIC INVESTIGATION REPORT

Suspicious Internal Reconnaissance Activity

Field	Value
Investigation ID	INC-2026-001
Document Status	Completed
Analyst	SOC Analyst (Lab Environment)
Investigation Date	30 August 2026
Primary Tool	Wireshark
Capture File	investigation.pcapng
Severity	MEDIUM
Environment	Lab / Simulated Environment

Prepared for cybersecurity learning and SOC portfolio documentation

1. Executive Summary

A network traffic investigation was performed on packet capture investigation.pcapng collected from a monitored workstation in a lab environment. The analysis first established a baseline of normal network communication, including a successful TCP 3-way handshake to a remote HTTPS service.

Later in the capture, the workstation 192.168.100.32 generated multiple outbound TCP SYN connection attempts toward 192.168.100.1 across destination ports 21, 22, 23, and 25. The sequence of connection attempts across multiple ports within a short period is indicative of potential internal network reconnaissance or port-scanning behavior.

A DNS query for the synthetic/non-standard domain suspicious-recon-domain-test.local was also observed and returned an NXDomain response. This DNS event provides supporting context but does not independently establish malicious activity.

No evidence of successful exploitation or data exfiltration was identified within the analyzed traffic. Additional endpoint and log evidence would be required to identify the responsible process or user and determine whether the activity was authorized.

2. Investigation Objectives

The investigation was conducted to:

- Identify the primary hosts involved in the observed network communication.
- Analyze TCP connection establishment and distinguish normal handshakes from abnormal attempts.
- Review DNS requests and responses associated with the monitored workstation.
- Identify unusual destination-port activity.
- Determine whether the traffic pattern is consistent with reconnaissance or port scanning.
- Construct a chronological incident timeline.
- Document indicators of interest and recommend appropriate SOC follow-up actions.

3. Environment and Evidence

Attribute	Observed Value	Description
Source Host	192.168.100.32	Monitored workstation / client
Target Host	192.168.100.1	Local gateway / target host in the lab
Observed Remote Host	18.154.7.97	Remote host observed during HTTPS communication
Associated Domain	search.brave.com	Domain associated with the observed HTTPS traffic
DNS Server	192.168.100.1	DNS server observed in the capture
Target Ports	21, 22, 23, 25	Ports targeted during suspicious connection attempts
Protocols	TCP, DNS, TLSv1.3	Primary protocols analyzed
Evidence	investigation.pcapng	Primary packet capture

Evidence scope: this report is based on the supplied packet capture. Statements about endpoint processes, users, authorization, exploitation, or compromise are not treated as confirmed unless supported by additional evidence.

4. Methodology

The packet capture was reviewed using a network-investigation workflow:

1. Establish a normal-traffic baseline.
2. Identify source and destination hosts and relevant protocols.
3. Inspect TCP streams and validate the 3-way handshake.
4. Review DNS requests and corresponding responses.
5. Filter TCP SYN packets to identify connection attempts.
6. Compare destination ports and timestamps to identify scanning-like patterns.
7. Build a timeline from relevant frames and events.
8. Assess findings, severity, and recommended response actions.

5. Network Analysis and Findings

5.1 Baseline and TCP 3-Way Handshake

Wireshark Display Filter:

tcp.stream == 4

Early traffic showed normal internet activity. Frame 212 contained the initial SYN, Frame 216 contained the SYN/ACK response, and Frame 217 contained the final ACK.

Frame	Direction	TCP Flags	Interpretation
212	192.168.100.32:49185 → 18.154.7.97:443	SYN	Connection initiation
216	18.154.7.97:443 → 192.168.100.32:49185	SYN, ACK	Server accepted the connection
217	192.168.100.32:49185 → 18.154.7.97:443	ACK	Handshake completed

Following the completed handshake, TLSv1.3 encrypted communication was observed. This stream provides a baseline example of successful client-server TCP communication.

5.2 DNS Traffic Investigation

Wireshark Display Filter:

dns

At approximately 15:36:38, 192.168.100.32 queried DNS server 192.168.100.1 for suspicious-recon-domain-test.local. A and AAAA queries were observed.

Time	Source	DNS Server	Query	Response
15:36:38	192.168.100.32	192.168.100.1	suspicious-recon-domain-test.local (A/AAAA)	NXDomain

Frame 28588 confirmed the NXDomain response. Because the domain is described as a synthetic/non-standard test domain, the event is relevant to the lab scenario. NXDomain alone is not sufficient evidence of malicious behavior.

5.3 Reconnaissance and Port-Scanning Detection

Wireshark Display Filter:

tcp.flags.syn == 1 && tcp.flags.ack == 0

Starting at approximately 15:35:59, the workstation 192.168.100.32 generated standalone TCP SYN packets toward 192.168.100.1 on multiple destination ports.

Port	Common Service	Observed Activity
21	FTP	TCP SYN connection attempt
22	SSH	TCP SYN connection attempt
23	Telnet	TCP SYN connection attempt
25	SMTP	TCP SYN connection attempt

Observed pattern:

192.168.100.32 → 192.168.100.1:21, 22, 23, 25

Several packets were marked as TCP retransmissions. Retransmission indicates that an expected response was not observed within the expected interval; it does not, by itself, prove that a port is closed or filtered.

The combination of a single internal source, a common target, multiple destination ports, and repeated SYN attempts within a short period is indicative of potential internal reconnaissance or port-scanning activity.

6. Incident Timeline

Time	Source	Destination	Protocol	Event / Analysis
15:32:33	192.168.100.32	18.154.7.97:443	TCP	Successful TCP 3-way handshake
15:32:33	192.168.100.32	18.154.7.97:443	TLSv1.3	Encrypted application communication observed
15:35:59	192.168.100.32	192.168.100.1:21	TCP	Outbound SYN; reconnaissance-like activity begins
15:36:21	192.168.100.32	192.168.100.1:22	TCP	Outbound SYN to SSH port
15:36:38	192.168.100.32	192.168.100.1	DNS	Synthetic domain query; NXDomain response
15:36:42	192.168.100.32	192.168.100.1:23	TCP	Outbound SYN to Telnet port
15:36:43	192.168.100.32	192.168.100.1:25	TCP	Outbound SYN to SMTP port

7. Indicators of Interest

Type	Indicator	Context
Source IP	192.168.100.32	Workstation generating the observed activity
Target IP	192.168.100.1	Internal target/gateway
Ports	21, 22, 23, 25	Multiple service ports targeted
Domain	suspicious-recon-domain-test.local	Synthetic/non-standard DNS query
Traffic Pattern	Repeated TCP SYN attempts	Consistent with possible reconnaissance

8. Findings

Finding	Assessment	Confidence
Normal HTTPS connection	Successful TCP handshake and TLSv1.3 traffic observed	High
Synthetic DNS activity	NXDomain response for non-standard test domain	High
Multiple-port connection attempts	Single internal host contacted several destination ports	High
Potential port scanning	Pattern is indicative of internal reconnaissance/port scanning	Medium-High
Successful exploitation	No evidence identified in analyzed traffic	Not observed
Data exfiltration	No evidence identified in analyzed traffic	Not observed

9. Severity Assessment

Severity: MEDIUM

The observed activity is suspicious because a workstation generated multiple connection attempts against different service ports on an internal host. Internal reconnaissance can precede further attack activity such as service exploitation or lateral movement.

The severity remains MEDIUM because the available PCAP does not show confirmed exploitation, credential compromise, malware execution, lateral movement, or data exfiltration.

10. Recommended SOC Response

10.1 Endpoint Investigation

- Inspect running processes and active network connections on 192.168.100.32.
- Review process execution history and command-line activity around 15:35:59.
- Review PowerShell, Windows Security, and other relevant endpoint logs.
- Use EDR telemetry, if available, to identify the process responsible for the connection attempts.

10.2 Log Correlation

- Correlate network timestamps with endpoint telemetry.
- If Sysmon is deployed, review Event ID 3 (Network Connection) for the relevant time window.
- Correlate DNS, firewall, and authentication logs with the network timeline.

10.3 Network Investigation

- Review traffic immediately before and after the detected scanning-like activity.
- Check whether additional internal hosts or ports were targeted.
- Look for subsequent successful connections to any of the identified services.

10.4 Containment

If the activity is confirmed to be unauthorized or malicious, consider isolating 192.168.100.32 from the internal network while endpoint investigation is performed.

11. Investigation Limitations

- The investigation is based solely on the available packet capture.
- The PCAP does not identify the exact process that generated the network traffic.
- The PCAP does not establish which user initiated the activity.
- The PCAP does not establish whether the scanning activity was authorized.
- Absence of evidence in the capture does not prove that exploitation or exfiltration did not occur outside the captured traffic.

12. Final Conclusion

The investigation identified multiple TCP SYN connection attempts originating from 192.168.100.32 and targeting 192.168.100.1 across ports 21, 22, 23, and 25. The pattern is indicative of potential internal network reconnaissance or port scanning.

A synthetic DNS query for suspicious-recon-domain-test.local was also observed during the same general period and returned NXDomain. This event is supporting context rather than independent proof of malicious activity.

No evidence of successful exploitation or data exfiltration was identified within the analyzed packet capture. Further endpoint, EDR, firewall, DNS, and authentication log correlation is recommended to determine the responsible process/user and establish whether the activity was authorized.

Final Assessment: MEDIUM — Potential Internal Reconnaissance / Port-Scanning Activity

13. Evidence Checklist for Portfolio

Evidence Item	Status	Recommended Filename
PCAP / PCAPNG	Available	investigation.pcapng
TCP handshake screenshot	To attach	tcp-handshake.png
DNS analysis screenshot	To attach	dns-analysis.png
Port-scanning screenshot	To attach	port-scanning.png
Timeline screenshot/table	To attach	timeline.png
Investigation notes	Prepared	investigation.md
Incident report	Prepared	incident-report.docx